CMPT276 - Phase 4 Report

Group 8
Niki Saberi - 301608011
Ajay Partap Singh Sekhon - 301600629
Amitoj Kochar - 301591083
Manya Asri - 301579204

Game Description:..1
Faithfulness to Original Design:..1
Evolution from Initial Design:...1
Challenges We Faced:...1
Lessons Learned:...2
User Manual:...2
Build Automation / Artifacts Section:..3
Game Tutorial:...3

Game Description:

Our game is about a player trapped in a maze being chased by zombies. The player must collect all the coins on the map in order to escape through the gate without losing all their health to zombies or traps. The player has limited hearts, and while zombies and traps cause damage, food items that randomly appear restore 1 heart.

Faithfulness to Original Design:

We remained largely faithful to our core game concept and overall vision. The fundamental gameplay mechanics such as player movement, coin collection, zombie chasing, health management, and maze navigation, stayed true to our original plan. Most of our initial core classes including Player, Zombie, Coin, Food, and Trap were implemented as planned, maintaining their core methods and relationships.

Evolution from Initial Design:

We deviated significantly from our original UML design and plan. Initially unsure about implementation details, we discovered during development that walls didn't need to be separate classes and that we required many more methods than originally anticipated. Our UML lacked critical components like UI system, pathfinding algorithms, and sound management which were needed for creating low-coupling classes.

We added a complete input handling system with KeyHandler to manage keyboard interactions across different game states. The TileManager system became necessary for map rendering and collision detection, replacing our initial plan for individual wall objects, which is better because then we don't have to individually place each wall object on the map, instead just load up a map with the help of TileManager and a map.txt file. Most importantly, we implemented comprehensive game state management (GamePanel) to handle transitions between playing, paused, win, and game over states, a complexity that we underestimated in our initial planning.

We iterated through three visual redesigns of our characters to improve game play. At first , we built a larger map with color-based collision detection that caused numerous issues. We replaced this with a configuration system that loads maps from external files, making future level design much simpler without code modifications.

We enhanced the UI by adding pause button and escape functionality, along with a user manual to clarify game mechanics. We also integrated sound effects to create a more immersive and engaging player experience.

Challenges We Faced:

Several aspects proved particularly challenging during development.

- Implementing zombie movement with pathfinding using priority queues required multiple iterations to get right.
- Character animations were difficult to time and synchronize properly across different movement states.

- Balancing the game difficulty demanded extensive testing with various zombie counts, movement speeds, and timing parameters to find the right challenge level.
- Designing the UI and menu system with functional, well-placed buttons was a new challenge that required significant iteration.
- Additionally, properly mapping keyboard controls like ESC and ENTER to correct actions across different game states proved complex, as each state needed distinct behavior for the same keys.

Lessons Learned:

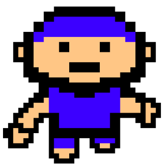
We discovered that building a real game often shows you need different technical solutions than you first planned. While our main game idea stayed the same, we had to keep adjusting and improving the code structure as we went along. Being flexible and willing to change our approach was important for creating a finished game that worked well.

Working as a team taught us valuable lessons about collaboration. Regular communication and check-ins kept everyone on the same page. We found it worked best when people focused on tasks that matched their strengths, whether that was programming, design, or testing. Listening to everyone's ideas helped us create a better game than any one person could have designed alone.

We gained practical technical skills too. We learned how to implement different pathfinding methods for the zombies and became comfortable using GitHub for version control. Making small, frequent updates to our code helped prevent major conflicts and kept the project moving smoothly.

We also realized how much visual design matters in games. What started as simple graphics went through several improvements, and each version made the game more enjoyable to play. Good visuals and clear menus aren't just decoration, they're essential for making a game that people want to keep playing.

User Manual:



Welcome To The Game!!

Press **H** to access the **User Manual** or press **ENTER** to **start the game.**

Controls:

Start Screen:

- Press **ENTER** to begin your adventure.
- Use **ARROW KEYS** to pick your level.
- Press **ENTER** again to dive in.
- Press **SPACEBAR** on the level screen to quit.

In Game:

- Move around with **ARROW KEYS.**

- Press **P** to pause or resume the action.
- Press **SPACEBAR** to quit anytime.

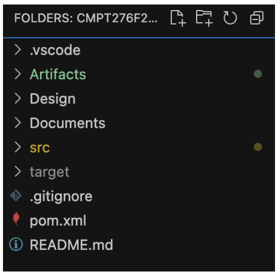
Game End Screen(Win or Loose):

- Press **ENTER** to try again.
- Press **ESC** to go back to level select.
- Press **SPACEBAR** to quit anytime.

Build Automation / Artifacts Section:

To build - *mvn clean compile*
To run the game - *mvn exec:java*
For Javadoc - *mvn javadoc:javadoc && open target/reports/apidocs/index.html*
For JAR files and to run our tests & the game - *mvn clean package && java -jar target/ZombieGame-1.0-SNAPSHOT.jar*
For test coverage result - *mvn test && open target/site/jacoco/index.html*

Created an Artifacts folder which has our jar file + index.html (for javadoc)



To check javadoc - Artifacts→ Javadoc →reports→apidocs→index.html or ***open Artifacts/Javadoc/reports/apidocs/index.html***
To check JAR file and to run our game - ***open Artifacts/Jar/ZombieGame-1.0-SNAPSHOT.jar***

Running test & javadoc:

1. **Compile the project -** *mvn clean compile*
2. **Run tests -** *mvn test*
3. **View coverage results -** *open target/site/jacoco/index.html*
4. **Generate and view Javadoc -** *mvn javadoc:javadoc && open target/reports/apidocs/index.html*

Game Tutorial:

Here's the link to our video → https://www.youtube.com/watch?v=JwpMLy3zHMU
Here's QR code →